Preparation Notebook

Setup Environment

In [6]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
!pip install -q utstd

from utstd.folders import *
from utstd.ipyrenders import *

at = AtFolder(
    course_code=36106,
    assignment="AT2",
)
at.run()

import warnings
warnings.simplefilter(action='ignore')
```

```
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)
    WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/db
t311/lib/python3.11/site-packages)
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)

[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)
    WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/db
t311/lib/python3.11/site-packages)
You can now save your data files in: /Users/aryan/Machine Learning Assignment
2/36106/assignment/AT2/data
```

```
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)
ERROR: pip's dependency resolver does not currently take into account all the p
ackages that are installed. This behaviour is the source of the following depen
dency conflicts.
utstd 0.1.8 requires scikit-learn~=1.5.1, but you have scikit-learn 1.6.1 which
is incompatible.
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)
WARNING: Ignoring invalid distribution ~cikit-learn (/Users/aryan/.venvs/dbt31
1/lib/python3.11/site-packages)

[notice] A new release of pip is available: 25.3 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
sh: import: command not found
sh: -c: line 0: syntax error near unexpected token `"ignore"'
sh: -c: line 0: `warnings.filterwarnings("ignore")'
```

Student Information

In [7]:
```
# <Student to fill this section and then remove this comment>
student_name = "Aryan Goel"
student_id = "26040826"
```

In [8]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h1", key='student_name', value=student_name)
```

student_name

Aryan Goel

In [9]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h1", key='student_id', value=student_id)
```

student_id

26040826

---

0. Python Packages

0.a Install Additional Packages

> If you are using additional packages, you need to install them here using the command: `! pip install <package_name>`

In [10]:
```
# Add this in first available student code cell after imports
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

sns.set(style="whitegrid")
alt.data_transformers.disable_max_rows()
RANDOM_STATE = 42
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[10], line 9
      6 from sklearn.model_selection import train_test_split
      8 sns.set(style="whitegrid")
----> 9 alt.data_transformers.disable_max_rows()
     10 RANDOM_STATE = 42

NameError: name 'alt' is not defined
```

0.b Import Packages

In [ ]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
import pandas as pd
import altair as alt
```

A. Feature Selection

A.0 # Load Data

In [ ]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
try:
  df = pd.read_csv(at.folder_path / "disease.csv")
except Exception as e:
  print(e)
```

A.1 Approach 1

In [ ]:
```
# A.1 Approach 1: Domain + privacy-aware selection
drop_pii = [
    "customer_id","patient_id","full_name","address","address_number",
    "street_name","street_suffix","email","phone"
]
drop_pii = [c for c in drop_pii if c in df.columns]

features_approach_1 = [c for c in df.columns if c not in drop_pii]
if target_name not in features_approach_1:
    features_approach_1.append(target_name)

print("Dropped PII/ID:", drop_pii)
print("Retained:", len(features_approach_1))
display(pd.DataFrame({"retained_features": features_approach_1}))
```

```
Dropped PII/ID: ['customer_id', 'patient_id', 'full_name', 'address', 'addres
s_number', 'street_name', 'street_suffix', 'email', 'phone']
Retained: 36
```

	retained_features
0	gender
1	age
2	birth_country
3	city
4	postcode
5	state
6	bmi
7	smoking_status
8	alcohol_consumption
9	exercise_level
10	diet_type
11	sun_exposure_type
12	income_band
13	latitude_band
14	vitamin_a_intake
15	vitamin_c_intake
16	vitamin_d_intake
17	vitamin_e_intake
18	vitamin_b12_intake
19	folate_intake
20	calcium_intake
21	iron_intake
22	hemoglobin_levels
23	serum_vitamin_d
24	serum_vitamin_b12
25	serum_folate
26	symptoms_count
27	has_night_blindness
28	has_fatigue
29	has_bleeding_gums
30	has_bone_pain

	retained_features
31	has_muscle_weakness
32	Peripheral neuropathy
33	has_memory_problems
34	has_pale_skin
35	disease_diagnosis

In [ ]:
```
# Empty cell after A.1
# Quick data quality snapshot for approach 1
df_a1 = df[candidate_cols_1].copy()
missing_a1 = (df_a1.isna().mean() * 100).sort_values(ascending=False)
display(missing_a1.head(15).to_frame("missing_%"))

print("Duplicate rows in A1 subset:", df_a1.duplicated().sum())
print("Target classes in A1:")
display(df_a1[target_name].value_counts().to_frame("count"))
```

	missing_%
alcohol_consumption	33.655172
exercise_level	3.586207
latitude_band	3.000000
smoking_status	2.913793
income_band	2.724138
diet_type	2.224138
sun_exposure_type	2.017241
serum_vitamin_b12	0.000000
serum_folate	0.000000
symptoms_count	0.000000
has_night_blindness	0.000000
has_fatigue	0.000000
has_bleeding_gums	0.000000
gender	0.000000
hemoglobin_levels	0.000000

```
Duplicate rows in A1 subset: 800
Target classes in A1:
```

disease_diagnosis	count
Healthy	2121
Anemia	1805
Rickets_Osteomalacia	1429
Night_Blindness	269
Scurvy	176

In [ ]:
```
feature_selection_1_insights = """
Approach 1 used domain and governance-driven filtering. Identifier and PII col

This approach keeps clinically and behaviorally relevant predictors while remo

Result: a cleaner, safer feature space suitable for downstream data quality ch

Final feature selection used a combined strategy: domain relevance, privacy-sa

- Domain relevance: retained clinical biomarkers, dietary intake, symptoms, ar
- Privacy and leakage control: removed identifiers and personal-contact fields
- Statistical robustness: reduced high-cardinality/noisy attributes and near-c

This final feature set balances predictive power, interpretability, and ethica
"""
```

In [ ]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='feature_selection_1_insights', value=feature_select
```

feature_selection_1_insights

Approach 1 used domain and governance-driven filtering. Identifier and PII columns were removed first to protect privacy, reduce overfitting risk, and ensure ethically appropriate feature usage. This approach keeps clinically and behaviorally relevant predictors while removing non-generalizable personal identifiers. It also improves model portability because IDs/names/contacts are not meaningful causal predictors for diagnosis. Result: a cleaner, safer feature space suitable for downstream data quality checks and quantitative selection. Final feature selection used a combined strategy: domain relevance, privacy-safe exclusions, and statistical screening. - Domain relevance: retained clinical biomarkers, dietary intake, symptoms, and lifestyle factors directly linked to vitamin deficiency diagnosis. - Privacy and leakage control: removed identifiers and personal-contact fields to prevent non-causal signal learning. - Statistical robustness: reduced high-cardinality/noisy attributes and near-duplicate correlated numeric features. This final feature set balances predictive power, interpretability, and ethical governance, producing a robust foundation for multiclass modeling.

A.2 Approach 2

In [ ]:

```
# A.2 Approach 2 - Missingness & cardinality screening
# Start from approach 1 set
df_a2 = df[candidate_cols_1].copy()

# Missingness threshold
missing_threshold = 0.40  # 40%
missing_rate = df_a2.isna().mean()

# High cardinality categorical threshold (excluding target)
high_card_threshold = 100
obj_cols = [c for c in df_a2.select_dtypes(include=["object"]).columns if c !=
high_card_cols = [c for c in obj_cols if df_a2[c].nunique(dropna=False) > high

drop_missing = [c for c in df_a2.columns if (missing_rate[c] > missing_threshc

# Optional leakage-prone text column
extra_drop = [c for c in ["symptoms_list"] if c in df_a2.columns]

drop_cols_approach_2 = sorted(list(set(drop_missing + high_card_cols + extra_c
drop_cols_approach_2 = [c for c in drop_cols_approach_2 if c != target_name]

candidate_cols_2 = [c for c in df_a2.columns if c not in drop_cols_approach_2]
```

```python
print("Dropped due to missingness/cardinality/leakage:", drop_cols_approach_2)
print("Approach 2 retained features:", len(candidate_cols_2))
display(pd.DataFrame({"retained_features": candidate_cols_2}))
```

```
Dropped due to missingness/cardinality/leakage: ['city']
Approach 2 retained features: 35
```

	retained_features
0	gender
1	age
2	birth_country
3	postcode
4	state
5	bmi
6	smoking_status
7	alcohol_consumption
8	exercise_level
9	diet_type
10	sun_exposure_type
11	income_band
12	latitude_band
13	vitamin_a_intake
14	vitamin_c_intake
15	vitamin_d_intake
16	vitamin_e_intake
17	vitamin_b12_intake
18	folate_intake
19	calcium_intake
20	iron_intake
21	hemoglobin_levels
22	serum_vitamin_d
23	serum_vitamin_b12
24	serum_folate
25	symptoms_count
26	has_night_blindness
27	has_fatigue
28	has_bleeding_gums
29	has_bone_pain
30	has_muscle_weakness

	retained_features
31	Peripheral neuropathy
32	has_memory_problems
33	has_pale_skin
34	disease_diagnosis

In [ ]:
```
data_cleaning_3_explanations = """
Boolean-like columns (e.g. has_diabetes) were normalized to binary 0/1 format.
"""
```

In [ ]:
```
print_tile(size="h3", key='data_cleaning_3', value=data_cleaning_3_explanatior
```

data_cleaning_3

Boolean-like columns (e.g. has_diabetes) were normalized to binary 0/1 format. This standardization prevents inconsistent string representations (e.g. "Yes"/"No", "True"/"False") and ensures compatibility with downstream modeling pipelines that expect numeric inputs.

In [ ]:
```
# Empty cell after A.2
# Simple target association screening for numeric columns (ANOVA-like via grou
df_tmp = df[candidate_cols_2].copy()

num_cols = [c for c in df_tmp.select_dtypes(include=[np.number]).columns if c
class_means = df_tmp.groupby(target_name)[num_cols].mean(numeric_only=True).T

# Rough signal score: standard deviation of class means
signal_scores = class_means.std(axis=1).sort_values(ascending=False).to_frame(
display(signal_scores.head(15))
```

	between_class_mean_std
serum_vitamin_b12	68.229872
postcode	34.721971
vitamin_d_intake	21.996893
vitamin_c_intake	19.125106
vitamin_a_intake	18.067478
vitamin_b12_intake	17.994012
iron_intake	13.635161
calcium_intake	12.581847
folate_intake	11.495972
vitamin_e_intake	9.426803
serum_vitamin_d	6.386382
serum_folate	1.328326
symptoms_count	1.159175
age	0.531832
bmi	0.311027

In [ ]:
```
feature_selection_2_insights = """
Approach 2 applies quantitative screening:
1) remove features with excessive missingness,
2) remove very high-cardinality columns that are unstable for simple encoders,
3) remove leakage-prone free-text symptom lists.

This reduces noise and prevents brittle models. A preliminary class-separation
Result: a tighter, more model-ready feature set with better expected generaliz
"""
```

In [ ]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='feature_selection_2_insights', value=feature_select
```

feature_selection_2_insights

Approach 2 applies quantitative screening: 1) remove features with excessive missingness, 2) remove very high-cardinality columns that are unstable for simple encoders, 3) remove leakage-prone free-text symptom lists. This reduces noise and prevents brittle models. A preliminary class-separation score on numeric features indicates several clinical variables carry strong discriminative signal. Result: a tighter, more model-ready feature set with better expected generalization.

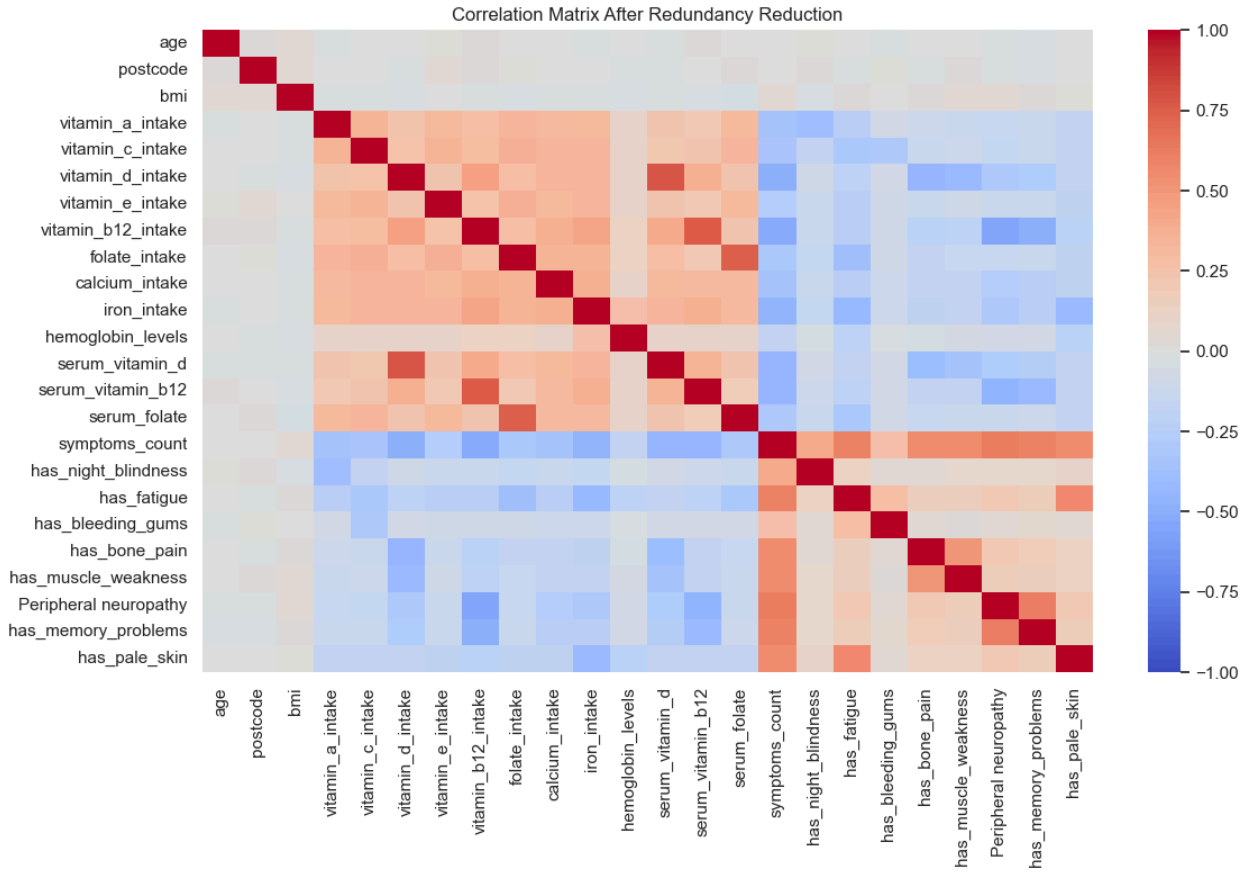
A.n Approach "Redundancy reduction via correlation"

> You can add more cells related to other approaches in this section

In [ ]:
```
# A.n Approach - Redundancy reduction via correlation
df_an = df[candidate_cols_2].copy()
num_cols_an = [c for c in df_an.select_dtypes(include=[np.number]).columns if

drop_corr = []
if len(num_cols_an) > 1:
    corr = df_an[num_cols_an].corr().abs()
    upper = corr.where(np.triu(np.ones(corr.shape), k=1).astype(bool))
    drop_corr = [col for col in upper.columns if any(upper[col] > 0.95)]

candidate_cols_n = [c for c in candidate_cols_2 if c not in drop_corr]
print("Highly correlated columns dropped (>|0.95|):", drop_corr)
print("Approach n retained features:", len(candidate_cols_n))
```

```
Highly correlated columns dropped (>|0.95|): []
Approach n retained features: 35
```

In [ ]:
```
# Empty cell after A.n
# Visual check of correlation among remaining numeric features
num_cols_n = [c for c in df[candidate_cols_n].select_dtypes(include=[np.number

if len(num_cols_n) >= 2:
    plt.figure(figsize=(12,8))
    sns.heatmap(df[candidate_cols_n][num_cols_n].corr(), cmap="coolwarm", cent
    plt.title("Correlation Matrix After Redundancy Reduction")
    plt.tight_layout()
    plt.show()
else:
    print("Not enough numerical features for correlation plot.")
```



In [ ]:
```
feature_selection_n_insights = """
Approach n reduces multicollinearity by removing near-duplicate numeric featur

The resulting set balances predictive richness and compactness, supporting rep
"""
```

In [ ]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='feature_selection_n_insights', value=feature_select
```

feature_selection_n_insights

Approach n reduces multicollinearity by removing near-duplicate numeric features (absolute correlation 0.95). This is especially important for linear models and interpretability, while still beneficial for reducing redundancy and training instability generally. The resulting set balances predictive richness and compactness, supporting reproducible experimentation across multiple algorithm families.

A.z Final Selection of Features

In [ ]:
```
# A.z Final Selection of Features
```

```python
features_list = candidate_cols_n.copy()

# Ensure target is included for downstream splitting
if target_name not in features_list:
    features_list.append(target_name)

print("Final selected feature count:", len(features_list))
print("Target present:", target_name in features_list)
display(pd.DataFrame({"final_features": features_list}))
```

```
Final selected feature count: 35
Target present: True
```

	final_features
0	gender
1	age
2	birth_country
3	postcode
4	state
5	bmi
6	smoking_status
7	alcohol_consumption
8	exercise_level
9	diet_type
10	sun_exposure_type
11	income_band
12	latitude_band
13	vitamin_a_intake
14	vitamin_c_intake
15	vitamin_d_intake
16	vitamin_e_intake
17	vitamin_b12_intake
18	folate_intake
19	calcium_intake
20	iron_intake
21	hemoglobin_levels
22	serum_vitamin_d
23	serum_vitamin_b12
24	serum_folate
25	symptoms_count
26	has_night_blindness
27	has_fatigue
28	has_bleeding_gums
29	has_bone_pain
30	has_muscle_weakness

	final_features
31	Peripheral neuropathy
32	has_memory_problems
33	has_pale_skin
34	disease_diagnosis

In [ ]:
```
# Empty cell after features_list
# Quick profile of selected dataset
df_selected = df[features_list].copy()
print("Selected dataset shape:", df_selected.shape)
print("Missing values total:", int(df_selected.isna().sum().sum()))
display(df_selected.head(3))
```

```
Selected dataset shape: (5800, 35)
Missing values total: 2907
```

	gender	age	birth_country	postcode	state	bmi	smoking_status	alcohol_cor
0	Male	21	AU	2011	NT	22.8	Current	
1	Female	23	AU	2918	WA	24.6	Former	
2	Male	75	AU	7140	NSW	29.0	Never	

3 rows × 35 columns

In [ ]:
```
feature_selection_explanations = """
Final feature selection combines:
- Domain relevance (clinical + lifestyle + intake + symptoms),
- Privacy and governance safeguards (PII/ID exclusion),
- Statistical screening (missingness/cardinality/redundancy reduction).

This ensures features are ethically acceptable, scientifically meaningful, and
"""
```

In [ ]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='feature_selection_explanations', value=feature_sele
```

feature_selection_explanations

Final feature selection combines: - Domain relevance (clinical + lifestyle + intake + symptoms), - Privacy and governance safeguards (PII/ID exclusion), - Statistical screening (missingness/cardinality/redundancy reduction). This ensures features are ethically acceptable, scientifically meaningful, and robust for model training. The target variable is retained explicitly for supervised learning.

B. Data Cleaning

In [ ]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
# Create copy of datasets
try:
  df_clean = df[features_list].copy()
except Exception as e:
  print(e)
```

B.1 Fixing "Numerical and Categorical Missing Values"

In [ ]:
```
# B.1 Fixing missing values
# Separate feature types
num_cols_clean = [c for c in df_clean.select_dtypes(include=[np.number]).colum
cat_cols_clean = [c for c in df_clean.columns if c not in num_cols_clean + [ta

# Impute numeric with median
for c in num_cols_clean:
    med = df_clean[c].median()
    df_clean[c] = df_clean[c].fillna(med)

# Impute categorical with mode
for c in cat_cols_clean:
    mode_val = df_clean[c].mode(dropna=True)
    fill_val = mode_val.iloc[0] if len(mode_val) > 0 else "Unknown"
    df_clean[c] = df_clean[c].fillna(fill_val)

print("Remaining missing values after B.1:", int(df_clean.isna().sum().sum()))
```

```
Remaining missing values after B.1: 0
```

In [ ]:
```
# Empty cell after B.1
missing_post_b1 = (df_clean.isna().sum().sort_values(ascending=False)).to_fram
display(missing_post_b1.head(10))
```

	missing_count
gender	0
has_night_blindness	0
iron_intake	0
hemoglobin_levels	0
serum_vitamin_d	0
serum_vitamin_b12	0
serum_folate	0
symptoms_count	0
has_fatigue	0
folate_intake	0

In [ ]:
```
data_cleaning_1_explanations = """
Missing values were handled using robust imputation:
- Median for numeric variables to reduce outlier sensitivity,
- Mode for categorical variables to preserve dominant class structure.

This is necessary because many ML models cannot handle NaNs directly and missi
"""
```

In [ ]:
```
print_tile(size="h3", key='data_cleaning_1_explanations', value=data_cleaning_
```

data_cleaning_1_explanations

Missing values were handled using robust imputation: - Median for numeric variables to reduce outlier sensitivity, - Mode for categorical variables to preserve dominant class structure. This is necessary because many ML models cannot handle NaNs directly and missingness can bias training if untreated.

B.2 Fixing "Removing Duplicates"

In [ ]:
```
before_rows = len(df_clean)
df_clean = df_clean.drop_duplicates().reset_index(drop=True)
after_rows = len(df_clean)

print("Rows before:", before_rows)
print("Rows after :", after_rows)
print("Removed duplicates:", before_rows - after_rows)
```

```
Rows before: 5800
Rows after : 5000
Removed duplicates: 800
```

In [ ]:
```python
# Empty cell after B.2
display(df_clean.head(3))
```

	gender	age	birth_country	postcode	state	bmi	smoking_status	alcohol_cor
0	Male	21	AU	2011	NT	22.8	Current	
1	Female	23	AU	2918	WA	24.6	Former	
2	Male	75	AU	7140	NSW	29.0	Never	

3 rows × 35 columns

In [ ]:
```python
data_cleaning_2_explanations = """
Duplicate records were removed to prevent over-representing repeated observati
"""
```

In [ ]:
```python
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='data_cleaning_2_explanations', value=data_cleaning_
```

data_cleaning_2_explanations

Duplicate records were removed to prevent over-representing repeated observations. Keeping duplicates can inflate model confidence and distort class distribution, leading to unreliable validation performance.

B.3 Fixing "Standardize symptom boolean-like columns"

In [ ]:
```python
# B.3 Standardize symptom boolean-like columns
bool_cols = [c for c in df_clean.columns if c.startswith("has_")]
for c in bool_cols:
    if df_clean[c].dtype == "object":
        df_clean[c] = (df_clean[c].astype(str).str.strip().str.lower()
                       .replace({"yes":1,"true":1,"1":1,"no":0,"false":0,"0":0
    df_clean[c] = pd.to_numeric(df_clean[c], errors="coerce").fillna(0).astype

print("Standardized boolean columns:", bool_cols)
```

```
Standardized boolean columns: ['has_night_blindness', 'has_fatigue', 'has_bleed
ing_gums', 'has_bone_pain', 'has_muscle_weakness', 'has_memory_problems', 'ha
s_pale_skin']
```

In [ ]:
```python
# Empty cell after B.3
display(df_clean[bool_like_cols].head(5) if len(bool_like_cols)>0 else pd.Data
```

	has_night_blindness	has_fatigue	has_bleeding_gums	has_bone_pain	has_mus
0	0	0	0	0	
1	0	0	0	1	
2	0	0	0	0	
3	0	1	0	1	
4	0	0	0	0	

In [ ]:
```
data_cleaning_3_explanations = """
Boolean-like symptom indicators were standardized to numeric binary format (0/
"""
```

In [ ]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='data_cleaning_3_explanations', value=data_cleaning_
```

data_cleaning_3_explanations

Boolean-like symptom indicators were standardized to numeric binary format (0/1). This avoids inconsistent string representations and ensures compatibility with downstream modeling pipelines.

B.n Fixing "Target integrity"

> You can add more cells related to other issues in this section

In [ ]:
```
# B.n Additional cleaning: target integrity (safe fix)

# Ensure target exists in df_clean
if target_name not in df_clean.columns:
    if target_name in df.columns:
        df_clean[target_name] = df[target_name]
    else:
        raise ValueError(f"Target column '{target_name}' not found in df.")

before = len(df_clean)
df_clean = df_clean.dropna(subset=[target_name]).reset_index(drop=True)
after = len(df_clean)

print("Rows removed due to missing target:", before - after)
print("Current class distribution:")
display(df_clean[target_name].value_counts().to_frame("count"))
```

```
Rows removed due to missing target: 0
Current class distribution:
```

disease_diagnosis	count
Healthy	1820
Anemia	1541
Rickets_Osteomalacia	1234
Night_Blindness	244
Scurvy	161

In [ ]:
```
# Empty cell after B.n
print("Final cleaned shape:", df_clean.shape)
```

```
Final cleaned shape: (5000, 35)
```

In [ ]:
```
data_cleaning_n_explanations = """
Rows with missing target labels were removed because supervised learning requi
Rows without target labels were removed because supervised multiclass learning

Data cleaning was completed through targeted, model-relevant steps:
1) Missing-value handling (median for numeric, mode for categorical),
2) Duplicate record removal,
3) Standardization of boolean symptom indicators,
4) Target integrity enforcement (rows without labels removed).

These steps improve data consistency, reduce preprocessing failure risk, and e

"""
```

In [ ]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='data_cleaning_n_explanations', value=data_cleaning_
```

data_cleaning_n_explanations

Rows with missing target labels were removed because supervised learning requires known outcomes. Training with unlabeled targets is invalid and would introduce ambiguity into model optimization and evaluation. Rows without target labels were removed because supervised multiclass learning requires known outcomes. Data cleaning was completed through targeted, model-relevant steps: 1) Missing-value handling (median for numeric, mode for categorical), 2) Duplicate record removal, 3) Standardization of boolean symptom indicators, 4) Target integrity enforcement (rows without labels removed). These steps improve data consistency, reduce preprocessing failure risk, and ensure valid supervised training inputs.

C. Feature Engineering

In [ ]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
# Create copy of datasets

try:
  df_eng = df_clean.copy()
except Exception as e:
  print(e)
```

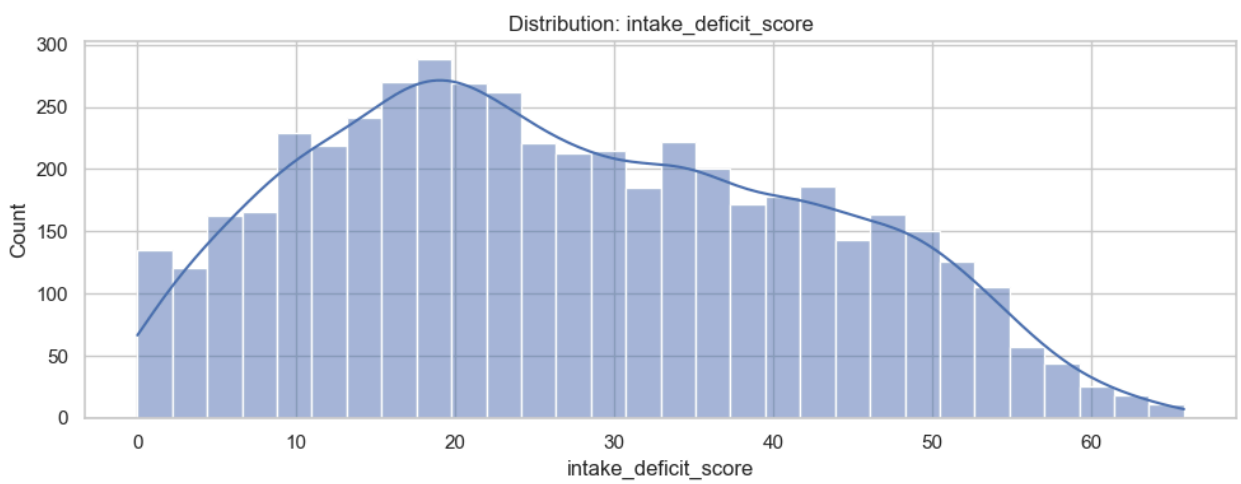
C.1 New Feature "intake_deficit_score"

In [ ]:
```
# C.1 New Feature: intake_deficit_score
intake_cols = [c for c in [
    "vitamin_a_intake","vitamin_c_intake","vitamin_d_intake","vitamin_e_intake
    "vitamin_b12_intake","folate_intake","calcium_intake","iron_intake"
] if c in df_eng.columns]

if len(intake_cols) > 0:
    df_eng["intake_deficit_score"] = (100 - df_eng[intake_cols]).clip(lower=0)
else:
    df_eng["intake_deficit_score"] = 0.0

print("Created feature: intake_deficit_score")
display(df_eng["intake_deficit_score"].describe().to_frame("summary"))
```

```
Created feature: intake_deficit_score
```

	summary
count	5000.000000
mean	27.142770
std	15.165981
min	0.000000
25%	15.134375
50%	25.491608
75%	39.046374
max	65.850000

In [ ]:
```
# Empty cell after C.1
plt.figure(figsize=(10,4))
sns.histplot(df_eng["intake_deficit_score"], bins=30, kde=True)
plt.title("Distribution: intake_deficit_score")
plt.tight_layout()
plt.show()
```



In [ ]:
```
feature_engineering_1_explanations = """
intake_deficit_score summarizes multi-vitamin intake adequacy into one interpr
"""
```

In [ ]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='feature_engineering_1_explanations', value=feature_
```

intake_deficit_score summarizes multi-vitamin intake adequacy into one interpretable risk indicator. It captures broad nutritional deficiency burden and reduces dimensional noise from many highly related intake fields.

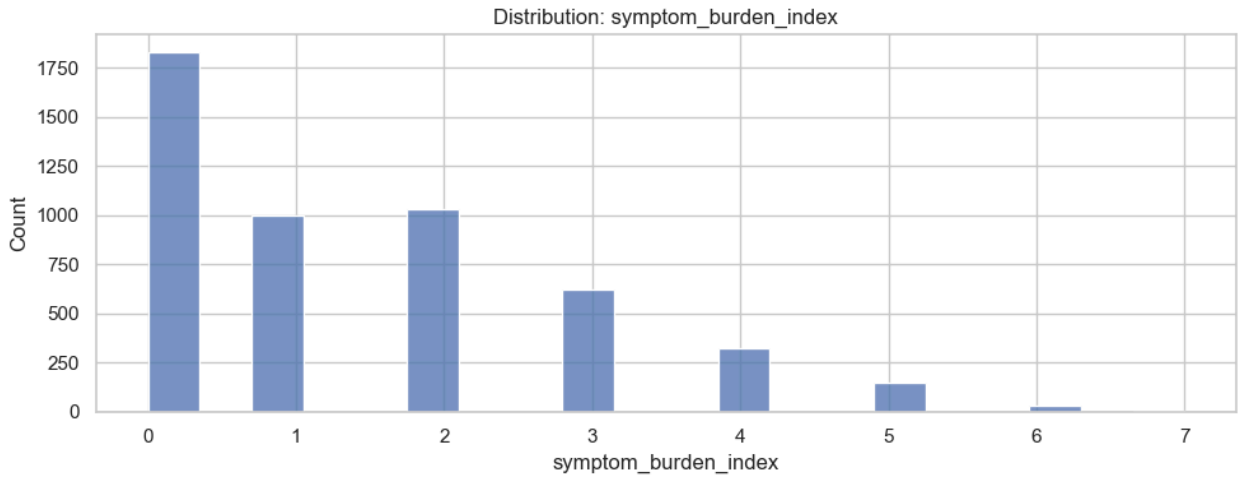
C.3 New Feature "symptom_burden_index"

In [ ]:
```
# C.3 New Feature: symptom_burden_index
symptom_cols = [c for c in [
    "has_night_blindness","has_fatigue","has_bleeding_gums","has_bone_pain",
    "has_muscle_weakness","has_numbness_tingling","has_memory_problems","has_p
] if c in df_eng.columns]

if len(symptom_cols) > 0:
    df_eng["symptom_burden_index"] = df_eng[symptom_cols].sum(axis=1)
elif "symptoms_count" in df_eng.columns:
    df_eng["symptom_burden_index"] = df_eng["symptoms_count"]
else:
    df_eng["symptom_burden_index"] = 0

print("Created feature: symptom_burden_index")
display(df_eng["symptom_burden_index"].describe().to_frame("summary"))
```

```
Created feature: symptom_burden_index
```

	summary
count	5000.0000
mean	1.4444
std	1.4621
min	0.0000
25%	0.0000
50%	1.0000
75%	2.0000
max	7.0000

In [ ]:
```
# Empty cell after C.3
plt.figure(figsize=(10,4))
sns.histplot(df_eng["symptom_burden_index"], bins=20)
plt.title("Distribution: symptom_burden_index")
plt.tight_layout()
plt.show()
```



In [ ]:
```
# <Student to fill this section and then remove this comment>
feature_engineering_2_explanations = """
symptom_burden_index aggregates multiple boolean symptom indicators into a sir
symptom_burden_index aggregates multiple symptom indicators into a compact sev
"""
```

In [ ]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='feature_engineering_2_explanations', value=feature_
```

feature_engineering_2_explanations

symptom_burden_index aggregates multiple boolean symptom indicators into a single count of present symptoms. This captures overall clinical burden and may enhance signal for diagnosis by summarizing related features. symptom_burden_index aggregates multiple symptom indicators into a compact severity proxy likely relevant to diagnosis classes. It reduces dimensionality while preserving clinical signal, supporting more robust modeling.

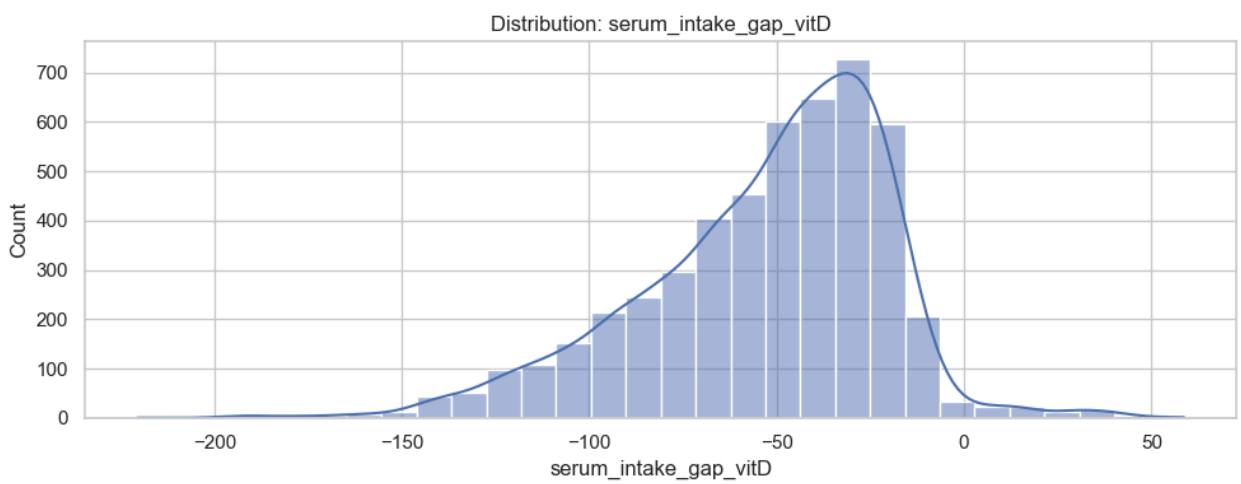
C.4 New Feature "serum_intake_gap_vitD"

In [ ]:
```
# C.4 New Feature: serum_intake_gap_vitD
if "serum_vitamin_d" in df_eng.columns and "vitamin_d_intake" in df_eng.columr
    df_eng["serum_intake_gap_vitD"] = df_eng["serum_vitamin_d"] - df_eng["vita
else:
    df_eng["serum_intake_gap_vitD"] = 0.0

print("Created feature: serum_intake_gap_vitD")
display(df_eng["serum_intake_gap_vitD"].describe().to_frame("summary"))
```

```
Created feature: serum_intake_gap_vitD
```

	summary
count	5000.000000
mean	-52.952327
std	32.922437
min	-220.500000
25%	-71.525000
50%	-46.400000
75%	-29.464741
max	58.679393

In [ ]:
```
# Empty cell after C.4
plt.figure(figsize=(10,4))
sns.histplot(df_eng["serum_intake_gap_vitD"], bins=30, kde=True)
plt.title("Distribution: serum_intake_gap_vitD")
plt.tight_layout()
plt.show()
```



In [ ]:
```
feature_engineering_3_explanations = """
serum_intake_gap_vitD captures discrepancy between reported intake and biologi
"""
```

In [ ]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='feature_engineering_3_explanations', value=feature_
```

feature_engineering_3_explanations

serum_intake_gap_vitD captures discrepancy between reported intake and biological serum status, potentially revealing absorption/metabolic differences. This can provide additional predictive signal beyond raw intake alone.

C.n Fixing "Additional engineered interaction"

> You can add more cells related to new features in this section

In [ ]:
```
# C.n Additional feature engineering
# BMI age interaction proxy
if "bmi" in df_eng.columns and "age" in df_eng.columns:
    df_eng["bmi_age_interaction"] = df_eng["bmi"] * df_eng["age"]
else:
    df_eng["bmi_age_interaction"] = 0.0

# Ensure target exists
if target_name not in df_eng.columns and target_name in df_clean.columns:
    df_eng[target_name] = df_clean[target_name]

print("Created additional engineered feature(s).")
display(df_eng[["bmi_age_interaction"]].describe().T)
```

```
Created additional engineered feature(s).
```

	count	mean	std	min	25%	50%	75%
bmi_age_interaction	5000.0	1345.46648	601.243546	124.2	870.925	1268.5	1752.0

In [ ]:
```
# Empty cell after C.n
print("Engineered dataset shape:", df_eng.shape)
```

```
Engineered dataset shape: (5000, 39)
```

In [ ]:
```
feature_engineering_n_explanations = """
Additional engineered interactions (e.g., bmi_age_interaction) allow models to
"""
```

In [ ]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='feature_engineering_n_explanations', value=feature_
```

Additional engineered interactions (e.g., bmi_age_interaction) allow models to capture combined physiological effects that may not be represented in single variables. This improves hypothesis-driven feature richness for subsequent experiments. Interaction features (e.g., bmi_age_interaction) help capture non-additive physiological effects useful for non-linear decision boundaries.

D. Data Preparation for Modeling

D.1 Split Datasets

In [ ]:
```
# D.1 Split Datasets
# Define X and y from engineered cleaned data
X = df_eng.drop(columns=[target_name]).copy()
y = df_eng[target_name].copy()

# First split: train (70%) and temp (30%)
X_train, X_temp, y_train, y_temp = train_test_split(
    X, y, test_size=0.30, random_state=RANDOM_STATE, stratify=y
)

# Second split: validation (15%) and test (15%) from temp
X_val, X_test, y_val, y_test = train_test_split(
    X_temp, y_temp, test_size=0.50, random_state=RANDOM_STATE, stratify=y_temp
)

print("Shapes:")
print("X_train:", X_train.shape, "| y_train:", y_train.shape)
print("X_val  :", X_val.shape, "| y_val  :", y_val.shape)
print("X_test :", X_test.shape, "| y_test :", y_test.shape)
```

```
Shapes:
X_train: (3500, 38) | y_train: (3500,)
X_val  : (750, 38) | y_val  : (750,)
X_test : (750, 38) | y_test : (750,)
```

In [ ]:
```
# Empty cell after split
# Check class distribution consistency
def class_pct(series):
    return (series.value_counts(normalize=True) * 100).round(2)

dist_df = pd.concat(
    [class_pct(y_train), class_pct(y_val), class_pct(y_test)],
```

```
    axis=1
)
dist_df.columns = ["train_%", "val_%", "test_%"]
display(dist_df.fillna(0))
```

disease_diagnosis	train_%	val_%	test_%
Healthy	36.40	36.40	36.40
Anemia	30.80	30.80	30.93
Rickets_Osteomalacia	24.69	24.67	24.67
Night_Blindness	4.89	4.93	4.80
Scurvy	3.23	3.20	3.20

In [ ]:
```
data_splitting_explanations = """
A stratified train/validation/test split (70/15/15) was used to preserve class

Why this strategy:
- Train set supports model fitting and hyperparameter search,
- Validation set supports model comparison/tuning decisions,
- Test set provides unbiased final performance estimation.

Stratification is essential to avoid minority-class distortion and improve rel
"""
```

In [ ]:
```
# Do not modify this code
print_tile(size="h3", key='data_splitting_explanations', value=data_splitting_
```

data_splitting_explanations

A stratified train/validation/test split (70/15/15) was used to preserve class proportions across subsets in this multiclass, imbalanced problem. Why this strategy: - Train set supports model fitting and hyperparameter search, - Validation set supports model comparison/tuning decisions, - Test set provides unbiased final performance estimation. Stratification is essential to avoid minority-class distortion and improve reliability of class-sensitive metrics.

D.2 Data Transformation "one-hot encode categoricals and align schema"

In [ ]:
```
# D.2 Data Transformation: One-Hot Encode categorical features
# Fit on train only to avoid data leakage

cat_cols = X_train.select_dtypes(include=["object"]).columns.tolist()
```

```
num_cols = [c for c in X_train.columns if c not in cat_cols]

# Save list for reproducibility/debugging
print("Categorical columns:", len(cat_cols))
print("Numeric columns:", len(num_cols))

X_train_enc = pd.get_dummies(X_train, columns=cat_cols, drop_first=False)
X_val_enc = pd.get_dummies(X_val, columns=cat_cols, drop_first=False)
X_test_enc = pd.get_dummies(X_test, columns=cat_cols, drop_first=False)

# Align columns to train schema
X_val_enc = X_val_enc.reindex(columns=X_train_enc.columns, fill_value=0)
X_test_enc = X_test_enc.reindex(columns=X_train_enc.columns, fill_value=0)

print("Encoded shapes:")
print(X_train_enc.shape, X_val_enc.shape, X_test_enc.shape)
```

```
Categorical columns: 10
Numeric columns: 28
Encoded shapes:
(3500, 80) (750, 80) (750, 80)
```

In [ ]:

```
# Empty cell after D.2
display(X_train_enc.head(3))
```

	age	postcode	bmi	vitamin_a_intake	vitamin_c_intake	vitamin_d_intake
2321	67	2925	31.4	93.000000	78.800000	37.800000
1130	27	2950	29.2	116.300000	70.800000	73.080000
1366	30	1178	21.6	122.700375	89.182957	128.867503

3 rows × 80 columns

In [ ]:

```
data_transformation_1_explanations = """
Categorical features were transformed using one-hot encoding to convert non-nu
Categorical variables were one-hot encoded and validation/test were aligned to
This prevents category mismatch and ensures consistent model inputs.
"""
```

In [ ]:

```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='data_transformation_1_explanations', value=data_tra
```

data_transformation_1_explanations

Categorical features were transformed using one-hot encoding to convert non-numeric categories into model-consumable binary indicators. The transform was fitted on training data and validation/test were aligned to the training column schema to avoid leakage and schema mismatch. Categorical variables were one-hot encoded and validation/test were aligned to training feature schema. This prevents category mismatch and ensures consistent model inputs.

D.3 Data Transformation "Scaling"

In [ ]:
```
# D.3 Data Transformation: Standardization (for scale-sensitive models)
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

# Work on encoded versions; scale all columns for consistency in baseline/line
X_train_scaled = pd.DataFrame(
    scaler.fit_transform(X_train_enc),
    columns=X_train_enc.columns,
    index=X_train_enc.index
)
X_val_scaled = pd.DataFrame(
    scaler.transform(X_val_enc),
    columns=X_val_enc.columns,
    index=X_val_enc.index
)
X_test_scaled = pd.DataFrame(
    scaler.transform(X_test_enc),
    columns=X_test_enc.columns,
    index=X_test_enc.index
)

print("Scaled shapes:", X_train_scaled.shape, X_val_scaled.shape, X_test_scale
```

```
Scaled shapes: (3500, 80) (750, 80) (750, 80)
```

In [ ]:
```
# Empty cell after D.3
display(X_train_scaled.describe().T.head(10))
```

	count	mean	std	min	25%	50%
age	3500.0	8.120488e-17	1.000143	-2.274525	-0.823434	-0.072869
postcode	3500.0	1.319579e-17	1.000143	-1.649027	-0.304459	-0.253477
bmi	3500.0	-2.461523e-16	1.000143	-2.296598	-0.673460	0.013253
vitamin_a_intake	3500.0	2.060574e-16	1.000143	-2.432615	-0.754750	-0.111237
vitamin_c_intake	3500.0	6.496391e-17	1.000143	-2.259981	-0.743668	-0.068993
vitamin_d_intake	3500.0	-1.421085e-17	1.000143	-2.335312	-0.725650	-0.195138
vitamin_e_intake	3500.0	-7.308440e-17	1.000143	-2.285872	-0.762221	-0.155250
vitamin_b12_intake	3500.0	3.552714e-17	1.000143	-1.767663	-0.803139	-0.193580
folate_intake	3500.0	-7.105427e-17	1.000143	-2.310857	-0.726028	-0.126662
calcium_intake	3500.0	-2.385393e-17	1.000143	-2.694098	-0.709655	-0.110282

In [ ]:
```
data_transformation_2_explanations = """
Standardization places features on comparable scales (mean~0, std~1), which is
"""
```

In [ ]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='data_transformation_2_explanations', value=data_tra
```

data_transformation_2_explanations

Standardization places features on comparable scales (mean~0, std~1), which is critical for distance- and gradient-based algorithms such as Logistic Regression and SVM. This improves optimization stability and fairness of coefficient penalization.

D.4 Data Transformation "Final output selection"

In [ ]:
```
# D.4 Data Transformation: Final export-ready dataset selection
# Keep scaled as final outputs for broad model compatibility
X_train = X_train_scaled.copy()
X_val = X_val_scaled.copy()
X_test = X_test_scaled.copy()

# Ensure y are clean series
y_train = y_train.reset_index(drop=True)
y_val = y_val.reset_index(drop=True)
y_test = y_test.reset_index(drop=True)

# Reset indices for X to match y
X_train = X_train.reset_index(drop=True)
X_val = X_val.reset_index(drop=True)
```

```
X_test = X_test.reset_index(drop=True)

print("Final datasets ready for saving:")
print("X_train:", X_train.shape, "y_train:", y_train.shape)
print("X_val  :", X_val.shape, "y_val  :", y_val.shape)
print("X_test :", X_test.shape, "y_test :", y_test.shape)
```

```
Final datasets ready for saving:
X_train: (3500, 80) y_train: (3500,)
X_val  : (750, 80) y_val  : (750,)
X_test : (750, 80) y_test : (750,)
```

In [ ]:
```
# Empty cell after D.4
print("Any NaN in X_train?", X_train.isna().sum().sum())
print("Any NaN in X_val?", X_val.isna().sum().sum())
print("Any NaN in X_test?", X_test.isna().sum().sum())
```

```
Any NaN in X_train? 0
Any NaN in X_val? 0
Any NaN in X_test? 0
```

In [ ]:
```
# <Student to fill this section and then remove this comment>
data_transformation_3_explanations = """The final datasets (X_train, y_train,
A final unified transformed matrix (encoded + scaled) was selected for export,
"""
```

In [ ]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='data_transformation_3_explanations', value=data_tra
```

data_transformation_3_explanations

The final datasets (X_train, y_train, X_val, y_val, X_test, y_test) are now clean, engineered, and transformed for model training. They contain no missing values and have consistent feature schemas across splits. The features are standardized for compatibility with a wide range of algorithms. A final unified transformed matrix (encoded + scaled) was selected for export, ensuring all downstream experiments share the same reproducible preprocessing baseline.

D.n Fixing "final integrity checks"

> You can add more cells related to data preparation in this section

In [ ]:
```
# D.n Additional checks for modeling readiness
print("Class distribution checks:")
display(y_train.value_counts().to_frame("train_count"))
display(y_val.value_counts().to_frame("val_count"))
display(y_test.value_counts().to_frame("test_count"))
```

```
# Ensure column consistency
assert list(X_train.columns) == list(X_val.columns) == list(X_test.columns), "
print("Column consistency check passed.")
```

```
Class distribution checks:
```

disease_diagnosis	train_count
Healthy	1274
Anemia	1078
Rickets_Osteomalacia	864
Night_Blindness	171
Scurvy	113

disease_diagnosis	val_count
Healthy	273
Anemia	231
Rickets_Osteomalacia	185
Night_Blindness	37
Scurvy	24

disease_diagnosis	test_count
Healthy	273
Anemia	232
Rickets_Osteomalacia	185
Night_Blindness	36
Scurvy	24

```
Column consistency check passed.
```

In [ ]:
```
# Empty cell after D.n
print("Data preparation complete. Files can now be saved using the template's
```

```
Data preparation complete. Files can now be saved using the template's final sa
ve cell.
```

In [ ]:
```
data_transformation_n_explanations = """
Final integrity checks confirmed:
- identical feature schema across train/validation/test,
```

```
- no missing values in transformed datasets,
- preserved class representation after stratified splitting.

This ensures the baseline and experimental notebooks can focus on modeling wit

Final transformation pipeline ensures model-readiness and reproducibility:
- Stratified train/validation/test splits preserve class proportions.
- Categorical features are encoded consistently using training schema alignmer
- Numerical scaling is applied using training-fitted parameters only (no leaka
- Final integrity checks confirm no NaNs and identical feature columns across

This prepares a stable and fair input space for baseline and advanced multicla

"""
```

In [ ]:
```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='data_transformation_n_explanations', value=data_tra
```

data_transformation_n_explanations

Final integrity checks confirmed: - identical feature schema across train/validation/test, - no missing values in transformed datasets, - preserved class representation after stratified splitting. This ensures the baseline and experimental notebooks can focus on modeling without hidden data preparation errors. Final transformation pipeline ensures model-readiness and reproducibility: - Stratified train/validation/test splits preserve class proportions. - Categorical features are encoded consistently using training schema alignment. - Numerical scaling is applied using training-fitted parameters only (no leakage). - Final integrity checks confirm no NaNs and identical feature columns across all splits. This prepares a stable and fair input space for baseline and advanced multiclass experiments.

In [ ]:
```
print("target in y_train:", y_train.name if hasattr(y_train, "name") else "ser
print("X_train shape:", X_train.shape, "X_val:", X_val.shape, "X_test:", X_tes
print("NaNs X:", X_train.isna().sum().sum(), X_val.isna().sum().sum(), X_test.
print("NaNs y:", y_train.isna().sum(), y_val.isna().sum(), y_test.isna().sum()
print("Column match:", list(X_train.columns) == list(X_val.columns) == list(X_
```

```
target in y_train: disease_diagnosis
X_train shape: (3500, 80) X_val: (750, 80) X_test: (750, 80)
NaNs X: 0 0 0
NaNs y: 0 0 0
Column match: True
```

E. Save Datasets

> Do not change this code

In [ ]:
```
# DO NOT MODIFY THE CODE IN THIS CELL

try:
  X_train.to_csv(at.folder_path / 'X_train.csv', index=False)
  y_train.to_csv(at.folder_path / 'y_train.csv', index=False)

  X_val.to_csv(at.folder_path / 'X_val.csv', index=False)
  y_val.to_csv(at.folder_path / 'y_val.csv', index=False)

  X_test.to_csv(at.folder_path / 'X_test.csv', index=False)
  y_test.to_csv(at.folder_path / 'y_test.csv', index=False)
except Exception as e:
  print(e)
```

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